

SUMMER INTERNSHIP TRAINING PRESENTATION



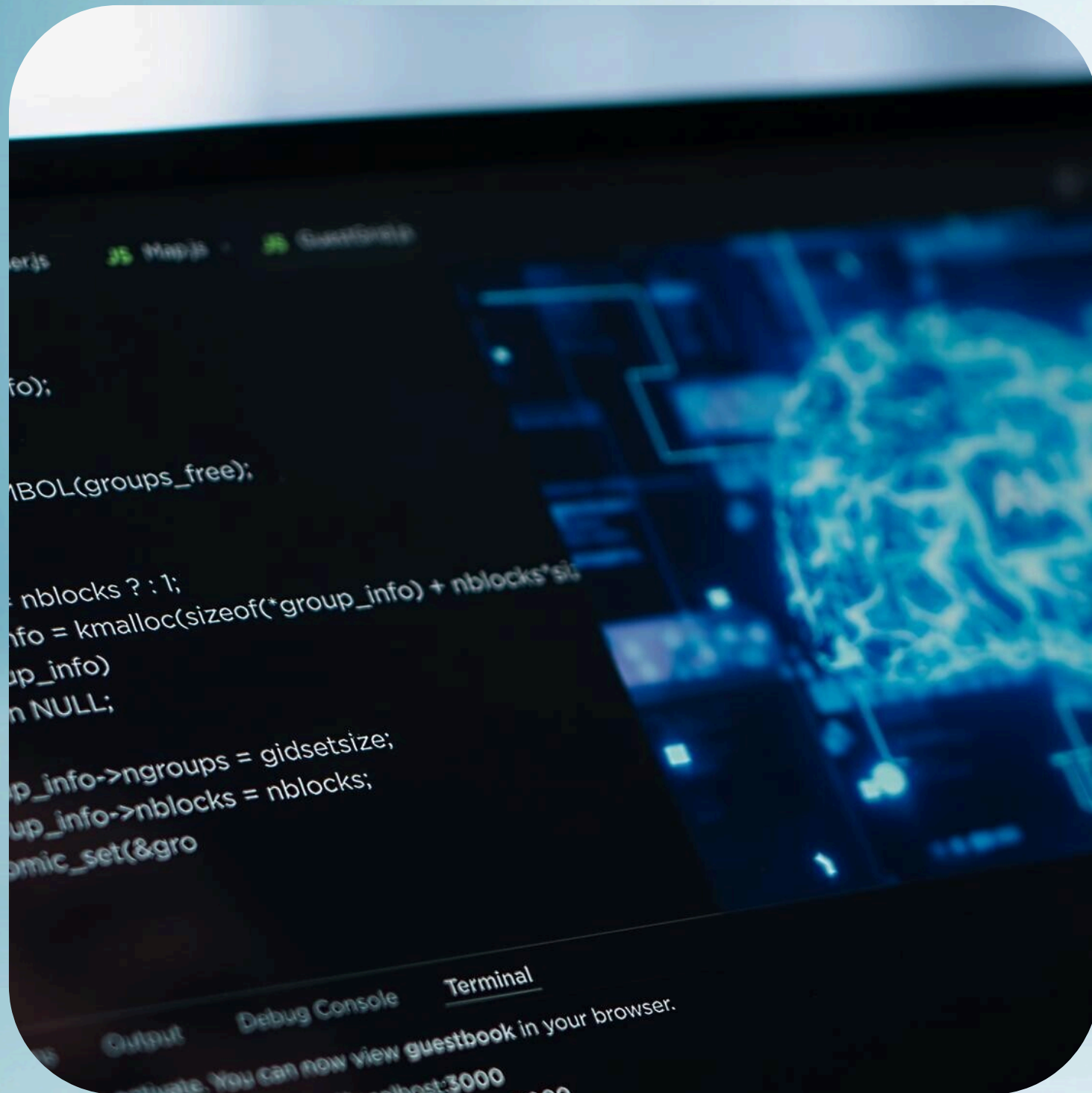
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**Topic: Data Structures & Algorithms (DSA) Using
Java + Project (Library Management System)**

Name: Satyam Yadav
College/University: Lovely Professional University
Date: 28/09/2025

**School of Computer Science and Engineering, Lovely Professional
University, Phagwara, India**

ABOUT THE TRAINING



- **Duration:** June 4, 2025, to July 15, 2025
Six weeks
- **Training Mode:** Hybrid (Online + Recorded Sessions)
- **Focus Areas:**
 - Java Fundamentals
 - Data Structures & Algorithms
 - Problem Solving & Competitive Programming
 - Mini Project Implementation (LMS)

CERTIFICATE



Certificate of Completion

This is to certify that

Satyam Yadav

studying at Lovely Professional University, has successfully completed training in **Data Structure and Algorithm using Java Programming** from CipherSchools during the period of **June 4, 2025, to July 15, 2025** .

The training comprised 70 hours of learning , and the participant's performance has been assessed as Satisfactory.

A handwritten signature in black ink, appearing to read "Anurag", is positioned above the name of the founder.

ANURAG MISHRA

Founder CipherSchools

CipherSchools, India

Certificate ID : CSW2025-15089



Scan to verify

TRAINING OBJECTIVES



- Learn **Java programming concepts** from basics to advanced
- Build a strong foundation in **DSA concepts**
- Solve real-world problems with **efficient algorithms**
- Gain exposure to **GitHub, LinkedIn, Resume building & Career guidance**
- Apply knowledge into **project development**

TRAINING SYLLABUS (WEEK-WISE)



- **Week 1:** Java Fundamentals & Complexity
 - Java basics, OOPs, Exception handling
 - Object class, Constructors
 - Time & Space Complexity, Big O, Θ , Ω
- **Week 2:** Arrays, Recursion, Strings, Searching & Sorting
 - Arrays & Recursion basics
 - String operations & interview questions
 - Searching (Linear, Binary)
 - Sorting (Bubble → Quick → Merge → Heap)



TRAINING SYLLABUS (WEEK-WISE)

- **Week 3-4:** Advanced Data Structures
 - Linked List (SLL, DLL, Circular)
 - Stacks & Queues (operations + applications)
 - Trees (BST, Traversals, Min/Max)
- **Week 5-6:** Advanced Topics
 - Hashing & Heaps
 - Graphs (BFS, DFS, Dijkstra, MST, Floyd-Warshall)
 - Dynamic Programming (Memoization & Tabulation)



ADDITIONAL ACTIVITIES



- **Git & GitHub Session** – version control, portfolio building
- **LinkedIn Session** – networking & job opportunities
- **Career Guidance** – roadmap for placements & higher studies
- **Resume Building** – ATS-friendly resume writing
- **Industrial Sessions** – experts from ISRO, NASA, Google, Microsoft
- **Tech Synergy (Coding Contest)** – competitive programming
- **Cipher Tank (Project Pitching)** – project pitching to experts

LEARNING OUTCOMES



- Solid understanding of Java + OOP
- Practical knowledge of DSA implementation
- Problem-solving skills with 150+ questions solved
- Experience in teamwork & coding contests
- Ability to design and implement projects

PROJECT INTRODUCTION



Library Management System

Books Users Transactions Borrow/Return Search

ID	Title	Author	Genre	Available
B001	The Great Gatsby	F. Scott Fitzgerald	Fiction	Yes
B004	Pride and Prejudice	Jane Austen	Romance	Yes
B005	The Hobbit	J.R.R. Tolkien	Fantasy	Yes
B002	To Kill a Mockingbird	Harper Lee	Fiction	Yes
B003	1984	George Orwell	Dystopian	Yes

Add New Book

Book ID:

Title:

Author:

Genre:

Add Book Remove Selected

- **Project Title:** Library Management System (LMS)
- **Objective:** Create an efficient system for managing books, users & transactions
- **Technology Used:**
 - Java (Core + Swing for GUI)
 - Data Structures: ArrayList, LinkedList, HashMap, PriorityQueue
- **Type:** Desktop Application

PROBLEM STATEMENT

- Manual library system → time-consuming & error-prone
- No quick way to:
 - Search for books
 - Track transactions
 - Maintain borrowing history
- Need for a **DSA-based automated system**



PROJECT GOALS



- Develop a **user-friendly GUI application**
- Implement **fast search & book management** using DSA
- Maintain **transaction history** efficiently
- Provide **analytics** (popular books/genres)
- Ensure **scalability** for future database integration

DATA STRUCTURES USED IN PROJECT



- HashMap → Store books & users with unique IDs
- ArrayList → Search results & dynamic lists
- LinkedList → Transaction records
- PriorityQueue → Find most popular genres quickly
- Stack/Queue concepts → For transaction flow handling

LMS FEATURES



-  Add / Remove Books & Users
-  Borrow / Return Books (with fine calculation)
-  Search by Title / ID / Genre
-  Maintain Transaction History
-  Display Most Popular Genres
-  Error Handling (invalid input, unavailable book)
-  GUI-based navigation

GUI DESIGN (JAVA SWING)



- **Tabbed Layout:**
 - Book Management
 - User Management
 - Borrow / Return
 - Transactions
 - Search
- **Tables:** display books, users, history
- **Forms & Popups:** for data entry & error messages
- **User-Friendly UI** → easy navigation

MAPPING DSA CONCEPTS



- **Arrays** → Store fixed book/user data
- **ArrayList** → Dynamic storage of borrowed books
- **HashMap** → Fast retrieval using Book/User ID
- **LinkedList** → Maintain continuous transaction logs
- **PriorityQueue** → Sort genres by popularity

PROJECT WORKFLOW



1. Admin/User logs in
2. Add/Remove/Search Books
3. Borrow/Return with updates in history
4. Transactions logged (LinkedList)
5. Popular books/genres updated in PriorityQueue
6. User-friendly reports generated

CHALLENGES FACED

- Designing **scalable GUI** with multiple features
- Implementing **error handling & validation**
- Mapping **theoretical DSA concepts** into real application
- Ensuring **smooth user experience**



OUTCOMES OF PROJECT



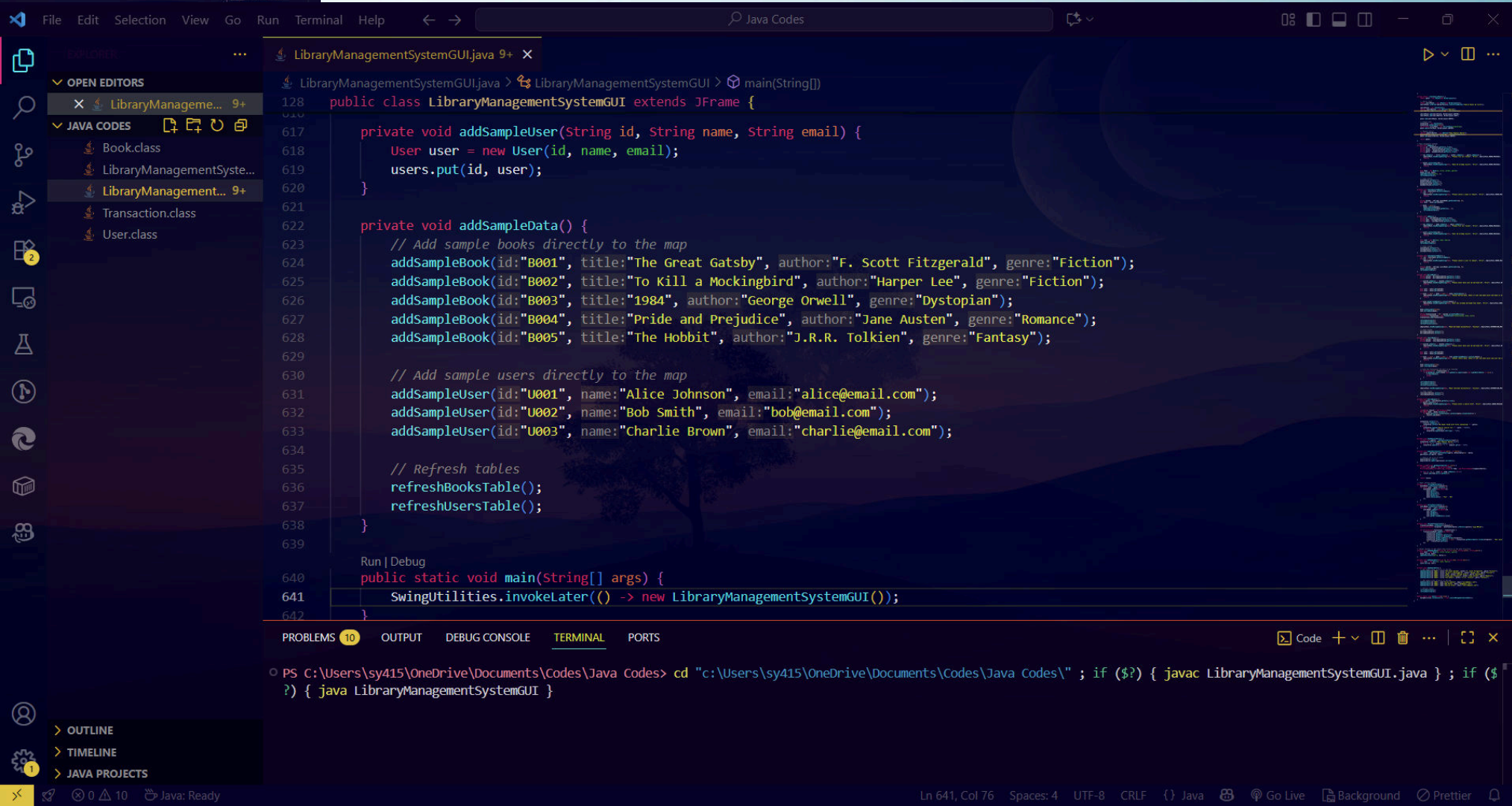
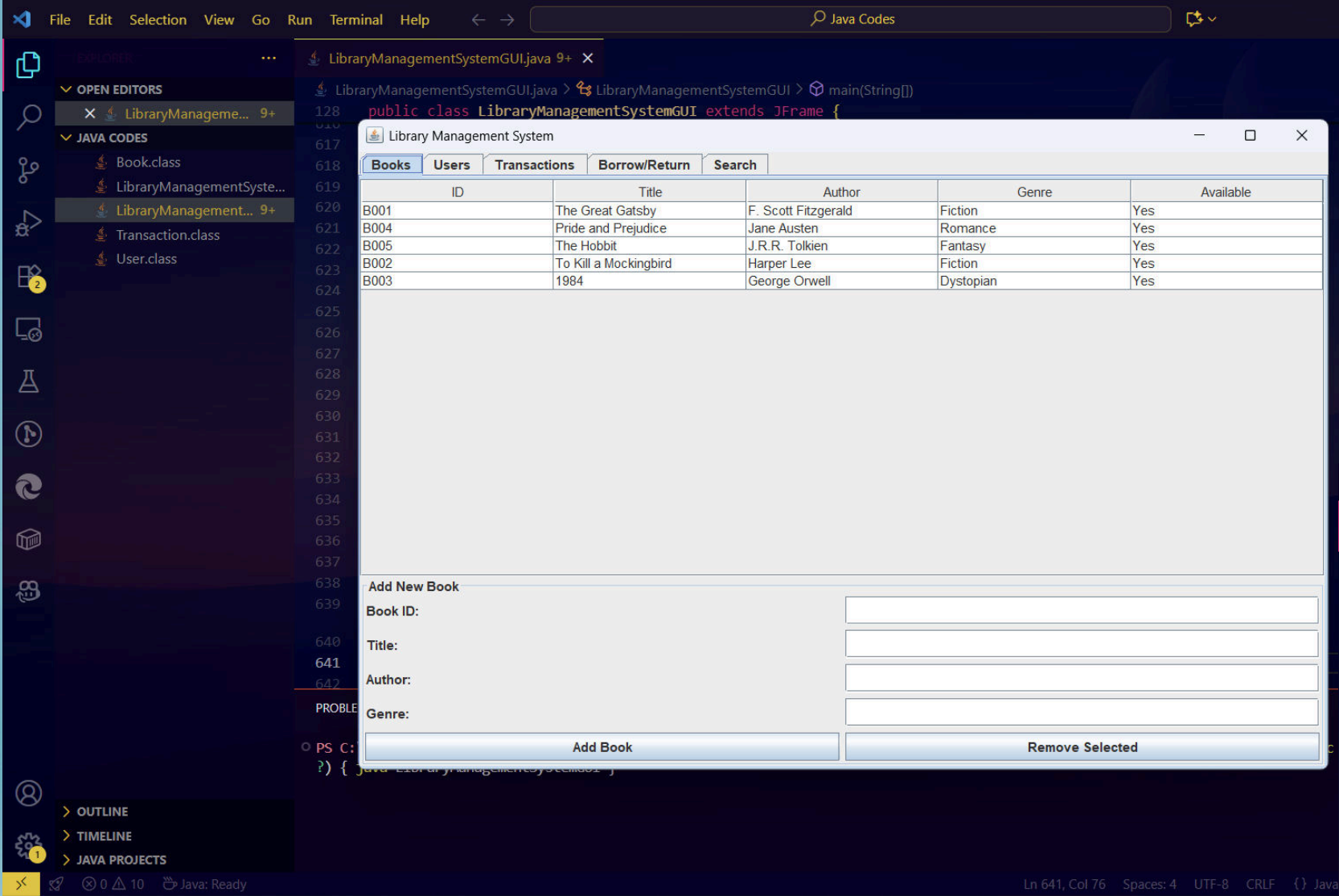
- Improved knowledge of Java Swing & DSA
- Built a **real-world** application from scratch
- Enhanced **problem-solving & debugging** skills
- Prepared for **placements & interviews**
- Project added to **resume & GitHub portfolio**

FUTURE SCOPE



- **Database Integration** (MySQL/PostgreSQL)
- **User Authentication & Roles** (Admin, Librarian, Student)
- **Cloud Deployment** for online access
- **Recommendation System** (suggest books based on usage)
- **Analytics Dashboard** for library insights

SCREENSHOTS



CONCLUSION / SUMMARY

- Training provided strong foundation in DSA using Java
- Learned both theoretical concepts & practical coding
- Applied knowledge in Library Management System project
- Developed technical + professional skills
- Ready for future challenges in software development



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Any Questions?